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Workbook · Energy efficiency

Pre-Solar Energy Checklist

Six steps to find what your building wastes before you size a solar PV or battery system. For owners and managers of commercial and industrial buildings in South Africa.

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How to use this workbook

Solar is usually sized on the bill your building pays today, and that bill includes everything the building wastes. This workbook helps you see that waste in about an hour, with information you already have.

Work through the six steps in order, filling in the tables by hand or in a copy of this PDF. By the end you'll know your night-time base load, when your demand peaks, which waste is easiest to remove, and what to ask any solar installer.

What you'll need

- Twelve months of electricity accounts (municipal or Eskom)
- Access to the main electricity meter, and to the building after hours once
- Thirty minutes with whoever runs the building day to day
- Any solar or battery quotes you've already received

1 Read your bill properly

Commercial tariffs charge for more than energy. Copy these figures from each monthly account. Not every tariff has every line; leave blanks where yours doesn't.

Month	Energy (kWh)	Peak demand (kVA)	Reactive energy (kVArh)	Total (R)

- [ ] Note your tariff name and whether it has time-of-use periods.
- [ ] If you pay reactive energy charges, your power factor needs attention. This is often one of the fastest paybacks.
- [ ] Note the month with the highest demand charge. You'll look for its cause in step 3.

2 Measure your night-time base load

The base load is what the building draws when it should be nearly empty. Solar can't supply it at night, and a battery that has to carry it is expensive. Read the main meter when the building closes and again when it opens.

Base load (kW) = kWh used overnight ÷ hours between the two readings

Reading	Date and time	Meter reading (kWh)
At closing		
At opening		
Hours between		kWh used:
Base load	kW	

Repeat it on a weekend. If your weekend base load is close to your weekday daytime load, equipment is running for nobody. A rule of thumb for offices and retail: A night-time base load above roughly a third of the daytime load deserves a closer look.

3 Find your demand peak

On a kVA demand tariff, the single highest half-hour of the month can set what you pay for capacity. Ask the people who run the site:

- [ ] What starts up first thing in the morning, and does it all start at once?
- [ ] Do large motors, compressors, chillers or ovens start together after a power cut?
- [ ] Is the peak before sunrise? If so, solar alone won't reduce it.
- [ ] Can start-ups be staggered by a few minutes without affecting operations?

> **Worth knowing**
>
> Your municipality or Eskom may be able to supply half-hourly interval data for your meter. It shows your real load profile, base load and peaks without any site measurement.

4 Walk the building for waste

Do this once during the day and once after hours. Tick what you find.

Air conditioning and ventilation

- [ ] Units running in empty rooms, at night or on weekends
- [ ] Heating and cooling running at the same time in the same space
- [ ] Setpoints colder than 23 °C in summer or warmer than 21 °C in winter
- [ ] Extraction fans running with no cooking, process or occupancy

Lighting

- [ ] Lights on in unoccupied areas, storerooms or car parks in daylight
- [ ] Old fluorescent or halogen fittings that could be LED
- [ ] No occupancy or daylight sensors in low-use areas

Compressed air, motors and pumps

- [ ] Audible air leaks, or the compressor running when production has stopped
- [ ] Compressor pressure set higher than the equipment needs
- [ ] Pumps or fans throttled with valves or dampers instead of speed control
- [ ] Motors running whether the process needs them or not

Refrigeration and kitchens

- [ ] Cold-room doors left open, damaged seals or iced-up evaporators
- [ ] Ovens, fryers or hot cupboards switched on hours before they're needed

The electrical installation

- [ ] Distribution boards, cables or transformers that feel hot to stand near
- [ ] Timers and controls nobody has checked since they were installed

5 Questions to ask your solar installer

- [ ] Was the system sized on measured load profiles, or on our bills?
- [ ] How much of our night-time base load will the battery have to carry, and what does that cost?
- [ ] When does our monthly maximum demand happen, and will this system reduce it?
- [ ] What is our power factor, and does the design address reactive energy charges?
- [ ] Is the inverter on the utility's approved list for our area?
- [ ] Who will commission the system and sign the SSEG report for the utility?
- [ ] What happens to the payback if we cut our base load by a quarter first?

6 Protect your tax benefit

The Section 12L energy efficiency tax incentive allows a deduction of 95 cents for every verified kWh saved, and the government has announced its extension to 31 December 2030. Savings must be measured and verified by a SANAS-accredited measurement and verification body against a baseline set **before** the efficiency project starts.

- [ ] Speak to your tax adviser before changing any plant or controls
- [ ] Arrange the baseline measurement first, then implement
- [ ] Keep the bills and readings from this workbook as supporting records

Want an engineer to do this properly?

An Energy Loss Audit measures your load at the incomer and main boards for one to two weeks, finds where the energy goes, and gives you a costed, ranked plan of savings measures, including how to size any solar system on the load that remains.

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This workbook gives general guidance only and is not a substitute for a site assessment. Facts were checked in October 2026; tariffs and incentives change, so confirm current figures with your utility and tax adviser. © 2026 Bynergex Consulting (Pty) Ltd.